

WEB3 FINANCE LLM:
CRYPTOCURRENCY MARKET
INSIGHTS & QUERY ANALYZER

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FACULTY OF INFORMATION SCIENCE & TECHNOLOGY

MULTIMEDIA UNIVERSITY

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BY

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ABSTRACT

The decentralised finance (DeFi) and the active growth of Web3 have formed huge financial ecosystems. However, the issue of deciphering on-chain and off-chain data remains a serious challenge to investors and analysts. Current analytics systems are complex and not user-friendly, which requires technical expertise making the use of real-time blockchain data inaccessible to non-professionals. To solve these issues, this research introduces the application of the Web3 Finance LLM: Cryptocurrency Market Insights and Query Analyzer, an analytical tool that provides interactions using large language models (LLMs) by a query-based system. It uses a Retrieval-Augmented Generation (RAG) system to put together dynamic blockchain information of networks such as Ethereum, Solana, and Polygon, with unstructured financial news and social media content. The proposed system can process this multi-source information to provide actionable insights, risk assessment, and visualizations by enabling the user to ask their questions in natural language. The prototype will democratize Web3 finance through simplified analysis of data, thus making it easier to use and encouraging innovation in DeFi applications.

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LIST OF ABBREVIATIONS/ SYMBOLS

AI	Artificial Intelligence
API	Application Programming Interface
BTC	Bitcoin
DeFi	Decentralized Finance
EMA	Exponential Moving Average
ERD	Entity Relationship Diagram
ETH	Ethereum
FIST	Faculty of Information Science and Technology
FYP	Final Year Project
LLM	Large Language Model
MAE	Mean Absolute Error
MATIC	Polygon Network Token
MMU	Multimedia University
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
RMSE	Root Mean Square Error
RSI	Relative Strength Index
SDLC	Software Development Life Cycle
SOL	Solana Network Token
TA	Solana Network Token
UI	User Interface
t	Time period or specific time step
P_t	Price of the cryptocurrency asset at time t
V_t	Trading volume of the asset at time t
y	Observed actual market value or trend direction
$\hat{y}$	Predicted market value or trend direction generated by the model
n	Total number of data observations in the dataset
Δ	Delta representing the change in a specific value
μ	Mean average of technical indicators
σ^2	Variance used to measure market volatility

Σ	Summation operator for calculating total values (e.g.,in RSI)
R^2	Coefficient of Determination for evaluating prediction accuracy

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CHAPTER 1

INTRODUCTION

1.1 Overview

Due to the rapid growth of Web3 and decentralized finance (DeFi), unprecedented opportunities have been made possible by the fact that users can now communicate with a blockchain-based financial system, e.g., in Ethereum or Solana or Polygon. However, the challenge of interpreting on-chain data, traditional financial data, and unstructured data is a significant challenge. This Final Year Project (FYP) proposes the implementation of Web3 Finance LLM: Cryptocurrency Market Insights and Query Analyzer, which is an LLM application that will supply such interactions with large language models and through a query-based interface. Users can pose questions in natural language and structured blockchain data (e.g. transaction histories, smart contract interactions) and unstructured data (e.g. news, social media) will be processed and provide actionable insights, risk assessment, and visualizations.

The foundation of this project is the intersection of AI and blockchain based on the recent literature (until December 2025) on the applicability of LLM in the cryptocurrency ecosystems. The inspirations include, but are not limited to, the framework trading agents that are run on LLM, vulnerability detection, sentiment analysis, and blockchain analytics. With access to Web3 finance democratized through the system, the complexity of data analysis will be reduced to promote innovation in DeFi applications.

1.2 Problem Statement

In the Web3 era, blockchain technology provides blockchain and financial data that are complex to navigate, which causes problems for the investor, the analyst, and the developer. These challenges include:

- **Data Complexity and Fragmentation:** The complexity and fragmentation of on-chain data from Ethereum, Solana, etc., is huge and complicated, requiring specific tools to analyze it. Integrating on-chain with off-chain, which is a financial dataset, also creates additional complexities.

- **Lack of User-Friendly Interfaces:** Most existing platforms like Dune Analytics require a lot of technical knowledge, making them difficult for non-technical users to use. This greatly reduces the use of real-time analysis from these platforms and keeps non-experts from utilizing on-chain data.
- **Insight Generation Gaps:** Traditional analytics tools do not provide conversational, context-based answers, assess risk properly, or provide good visualizations that result in a number of bad and inefficient decisions made in the highly volatile cryptocurrency markets.
- **Scalability and Accuracy Issues:** Advanced AI is needed to give complete and accurate results for DeFi query types like dynamic queries, which can either relate to market insights or anomaly detection.

Although investors, analysts and developers face similar issues today as they did prior to LLM's introduction, LLM has not yet been widely adopted as an integration tool for many domains, including Web3 finance, and consequently has limited opportunity for growth. This project therefore meets this gap through the creation of an intuitive LLM-driven analyzer.

1.3 Project Objectives

The goals of this FYP are set as follows, which align with the proposal:

- **Develop Core LLM Integration:**
- **Implement User Interface:**
- **Enable Multi-Source Data Processing:** Implement a dual-stream data processing logic that separates quantitative on-chain technical data for trend forecasting from qualitative off-chain text for contextual daily summaries.
- **Ensure Innovative Enhancements:** Enhance existing tools like Dune Analytics through conversational AI in the system for ease of use and deeper insights into cryptocurrency market analysis.

Innovative Improvements: Build on top of applications like Dune Analytics by adding capabilities in conversational AI for better usability and insight depth on the analysis

of cryptocurrency markets. These objectives are to be able to design a prototype that, through one practical application of Web3, will maneuver AI and blockchain.

1.4 Project Scope

The project will be concentrating its efforts on creating a prototype of the LLM-powered system that will be able to analyze the cryptocurrency market and perform data queries on Web3. The assigned duties will only be limited to a successful application that lets people work with blockchain and financial data via normal language inquiries. The system is aimed mainly at the large blockchain networks such as Ethereum, Solana, and Polygon because they are the most popular ones in the decentralized finance (DeFi) sector, and they also offer quite a lot of on-chain data to be analyzed.

The project will incorporate an LLM that will be responsible for interpreting queries and generating insights, will allow for the extraction of both on-chain blockchain data and off-chain financial data, and will include unstructured textual data like news articles or posts from social media that are related to cryptocurrencies. The system is going to give detailed insights, risk indicators, and easy visual representations for users to comprehend the market behavior. The usability, interpretability, and practical implementation are the main concerns of the project and not the comprehensive financial forecasting or automated trading execution.

Some of the activities that the project will not cover are live trading, managing portfolios, deploying smart contracts, or executing transactions directly on-chain. The system will not be issuing any financial advice or making any guarantees regarding investment returns either. Its role is limited to that of a research and development tool that confirms the usability and efficiency of LLMs in Web3 finance analytics applied under an academic setting.

The scope specifically focuses on utilizing historical 2025 technical indicators for market prediction to ensure model robustness. Unstructured data, such as news and social media, are utilized strictly for providing daily context and summaries to the user and do not serve as input features for the predictive engine to avoid volatility noise.

1.5 Report Organisation

The report consists of 5 chapters, and each chapter deals with one specific aspect of the Final Year Project. The first chapter introduces the project in terms of its background, problem statement, objectives, scope, and the main reason for the creation of an LLM-based Web3 finance analytics system. Besides, it presents the context and importance of the research.

Chapter 2 explores literature on Web3 finance, DeFi and LLMs with their uses in cryptocurrency analytics, trading, security, and sentiment analysis. It reveals the existing gaps in research and hence the need for the proposed solution.

In Chapter 3, the work plan is discussed regarding the applied methodology in this project, starting with the overall LLM-based approach, through system architecture, workflow design, and deployment considerations, to the use of Retrieval-Augmented Generation (RAG). This chapter demonstrates how the system is designed and set up to achieve the project objectives.

In Chapter 4, system implementation is described along with the discussion of the technologies, tools, and frameworks that were used to develop the prototype. It also elaborates on the implementation of the major system components such as data retrieval, LLM integration, user interface, and visualization modules.

The conclusion of Chapter 5 encompasses a summary of the project's success, a critical account of its drawbacks, and a suggestion for further studies concerning LLM-powered Web3 and DeFi analytics, thereby marking the end of the report.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Web3 Finance and LLM-Driven Analytic

Web3 is considered a paradigm shift in the digital infrastructure that will transition us from the read-only Web1 and interactive Web2 to a decentralized and user-centric digital infrastructure based on the blockchain technology (Buterin et al., 2013; Wood et al., 2014, as cited in Huang et al., 2025). At its simplest, Web3 finance, commonly referred to as Decentralized Finance (DeFi), makes use of permissionless blockchains to facilitate peer-to-peer financial services, such as lending, borrowing, trading and asset management, without the use of traditional intermediaries (Ozili, 2022, as cited in Huang et al., 2025). Key components include smart contracts which have self-executing codes on a platform such as Ethereum that helps to automate transactions with a level of transparency and security (Chatterjee and Ramamurthy, 2025; Geren et al., 2025, as cited in Huang et al., 2025). On-chain analytics, a critical aspect, consists in the analysis of the data stored on the blockchains (e.g. transaction records, wallet activities) with an intent to find insights in the market dynamics, user activities and risks (Geren et al., 2025, as cited in Huang et al., 2025). This availability of data, in combination with methods of off-chain data (such as social media) make this a very rich environment for predictive modelling.

The integrating Large Language Models (LLMs) in Web3 Finance is a major evolution in the field of analytics. Largely transformer-based language models (LM) (GPT-4, OpenAI et al., 2024, as cited in Guo et al., 2025; Llama, Touran et al., 2023, as cited in Guo et al., 2025), which have been trained on huge data bodies (also known as anywhere from T1B to T100B) have been demonstrating emergent abilities in reasoning, generation and multi-modal processing (DeepSeek-AI et al., 2025; Team et al., 2024, as cited in Guo et al., 2025). Their ability to handle both unstructured and structured on-chain data makes them suitable for blockchain as they solve the problems of standard machine learning like data shortage and interpretability (Wang et al., 2025a, as cited in Guo et al., 2025). For instance, LLMs facilitate the zero-shot learning that

allows the crypto markets to be tailored to volatile environments and not need complex retraining (Wahidur et al., 2023).

Previous work suggests that the data silos of Blockchain make it a limitation with regards to interoperability and LLMs suffer from hallucinations and computational requirements (Gong et al., 2024). Surveys emphasize bidirectional of synergies LLMs improving the security and analytics of blockchains, blockchain guarantees the data integrity of LLMs (He et al., 2024; Gong et al., 2024). This is the coming together that solves the gaps in traditional finance because of the lack of conflicting decentralized models, and the way is paving for smart and autonomous systems in the Web3.

2.2 LLMs in Cryptocurrency Trading and Market Prediction

The thing that predominantly characterizes the cryptocurrency market is its high volatility, which, together with the nature of the phenomena, makes it a non-linear market and thus the prediction becomes a quite difficult one (Drozdź et al., 2023, as cited in Li et al., 2024). Regression, ANN, and SVM are the traditional ML models that can uncover and reveal the underlying patterns in the past, but these models struggle a lot with the rapid changes in public sentiments (Adedigba et al., 2025; Amirzadeh et al., 2025). LLMs are based on reasoning, context comprehension, and multisensory processing; thus, they are giving the traders and forecasters more flexible and creative methods to deal with the future price. In demand forecasting, the initial sentiment techniques depending on dictionaries have been substituted with LLM-based contextual analysis; indeed, recent breakthroughs like Makri et al. (2025) indicate that few-shot LLMs have a lower MAE in the short-term prediction of Ethereum and they also outperform the baselines when the data is scarce. Other studies in the field of blockchain integration and social signals also show very high R^2 (>0.99) values (Gurgul et al., 2023), but still real-time adaptability is not that good. For trend classification, the accuracy levels reached by the ANN and SVM models are above 90% (Tanrikulu et al., 2024), and through data discretization the performance is improved by 10%, thus the importance of preprocessing is revealed where LLMs naturally dominate. The results of empirical studies have found that LLMs constantly provide an RMSE reduction of 15-20% compared to the ARIMA and SVR models (Chen, 2025), even though most of the other models still ignore the cross-asset interactions.

The zero-shot and few-shot LLM trading systems not only reduce the necessity for large historical datasets but also bring in other benefits as well. While the reflective LLM agents proposed by Li et al. (2024) claim 31% more profits through self-feedback methods, the zero-shot models accomplish a performance rose up to 40% after the fine-tuning (Wahidur et al., 2023). Few-shot learning techniques introduced by Makri et al. (2025) is giving good outcomes even with minimal data, but a major issue remains with hallucination in uncertain market situations. Conversely, sentiment-aware modeling creates a further prediction boost where multimodal LLMs that can do both textual and visual data processing reach an accuracy of 75% and being about 10% more effective than unimodal (Liu et al., 2025a), still the performance gets lower in bearish markets due to negative signal bias. Multiagent LLM trading frameworks (Jin et al, 2024; Singhi, 2025) not only support but also broaden the scope of decision-making thus leading to an increase in returns by up to 30% in the market's bullish phases. Nevertheless, these systems usually assume the presence of cooperative and honest agents, so the problems of trust, coordination and integrity in the decentralized trading environments remain unsolved.

Sentiment, aware strategies incorporating social signals as part of sentiment analysis in portfolio optimization have led to remarkable figures in terms of improvements. Haase et al. (2025) show that such strategies resulted in cumulative returns of 38% against 821% for regular benchmarks, albeit the presence of noisy unstructured text made the task difficult. Various ensemble approaches like Gradient Boosting and XGBoost are able to reach R of about 0.98 and are approximately 10% more accurate than LSTM (Adedigba et al., 2025), but they come with a higher computational cost.

On the other hand, large language models (LLMs) are capable of further improving allocation decisions by employing factsubjectivity reasoning (Wang et al., 2024) which leads to profit increments of 710%; nevertheless, stronger models sometimes can be found in a position of underperformance during volatile markets as they rely too much on factual priors rather than market intuition (Li et al., 2024). Overall, LLM-driven methods have been reported to increase the Sharpe ratios in the range of 20%, while the finetuning for particular domains is still considered of utmost importance.

2.2.1 Comparison to Traditional ML

LLMs surpass ML in adaptability (e.g., +20% Sharpe; Adedigba et al., 2025) but lag in speed/scalability. Gaps: realtime integration and crosschain generalization.

Aspect	Traditional ML (e.g., XGBoost)	LLMs (e.g., GPT-4)	Performance Gap
Price Accuracy	$R^2=0.85-0.95$ (Adedigba et al., 2025)	$R^2=0.98$ (Gurgul et al., 2023)	+5-10% (Adedigba et al., 2025; Gurgul et al., 2023)
Volatility Handling	Poor in bears (Amirzadeh et al., 2025)	Better via reasoning (Wang et al., 2024)	+15% returns (Wang et al., 2024)
Data Efficiency	High data needs	Few-shot (Makri et al., 2025)	Reduces data by 80% (Makri et al., 2025)

Table 2-1: *Comparison of LLM vs. Traditional ML in Crypto Trading*

Predictions of prices using experimental designs typically involve historical datasets such as four years of Ethereum data (Makri et al., 2025), and they use average absolute error (MAE) and root mean square error (RMSE) among others as their measures of quality. One of the common mistakes is data leakage that occurs when future data is used for training models unintentionally (Amirzadeh et al., 2025) which results in R (coefficient of determination, measuring variance explained; up to 0.99 in Gurgul et al., 2023) that is significantly higher than it should be. Look, ahead bias means the use of information that would not be available in the future and thus the results are biased (Tanrikulu et al., 2024). Usually, evaluation periods are very short (e.g., 1, 3 days; Makri et al., 2025) and therefore long, term volatility is not considered. The real labeling depends on price thresholds; thus it may be subjective (Adedigba et al., 2025).

Portfolio optimization strategies utilize rolling, window backtests (Chen, 2025) together with the Sharpe ratio for 1 and 2 years period respectively. One of the main drawbacks is that the models are tailored only to bull markets (Haase et al., 2025) and

transaction costs are ignored. The efficacy of multi-agent systems is proven on synthetic data (Singhi, 2025), but they do not experience real, world slippage.

2.3 LLMs for Smart Contract Auditing, DeFi Security, and Fraud Detection

Smart contracts are governed by immutability which means that once the contract has been deployed, the vulnerabilities cannot be fixed, this has been the reason for billions of dollars in financial losses in the DeFi ecosystems (DefiLlama, 2024, as cited in Mothukuri et al., 2024). The widely accepted static and dynamic analysis tools such as Mythril and Slither serve their purpose but most of the time they have high false-positive rates and provide little reasoning in the context when it comes to analyzing complicated logic (Kevin et al., 2025). LLMs have countered these limitations through the interpretation of code semantics, comments, and execution patterns allowing for auditing depending on the context that detects vulnerabilities of reentrancy, oracle manipulation, and improper access control. Previous studies indicate the capability of LLMs to achieve perfect recall (100%) in discovering reentrancy and access-control problems which is much higher than the 70% recall level that is generally associated with the static analysis methods (Kevin et al., 2025; Jie et al., 2025). The models that were fine-tuned have generated results that are even more impressive (Daspe et al. 2024) announcing F1 two measures of 10–15% increments in the benchmarking of Ethereum auditing. Moreover, combining agents that integrate code, comments and bytecode, precision is further increased by 16% (Jie et al., 2025). On the other hand, such improvement has not completely eradicated the difficulty associated with fresh forms of exploits that are not similar to any of the known patterns, hence, the need for continuous adaptation is emphasized.

The progress made in auditing based on Large Language Model (LLM) has come to the point where it is not just limited to the evaluation of a single model but also includes Retrieval-Augmented Generation (RAG) and multi-agent systems, which lead to the accuracy of ~70% in code reasoning and vulnerability coverage (Kevin et al., 2025; Jie et al., 2025). The smart contract code is not the only area where LLMs have a positive impact on the DeFi security as they can also spot price manipulation and fraud with an impressive up to 88% accuracy which is a perfect mix of on-chain monitoring

and model-based reasoning (Liu et al., 2025b; Ahmed et al., 2025). However, the situation is still not easy for the practical implementation due to the high computational requirement for edge devices and the need for real-time oracle data, hence the recommendation for optimized hybrid architectures that can adapt to changing adversarial scenarios (Karanjai et al., 2024).

2.3.1 Comparison to Static/Dynamic Analysis Tools

The use of LLMs gives rise to a great extent and accuracy in the meaning of words (70-90% F1 vs. 50-70% for Mythril; Kevin et al., 2025), but they still require a lot of computational power. Also, most of them are still not usable across different blockchains.

Tool Type	Precision/Recall	Limitations	Examples
Static (Mythril)	60-80% (Kevin et al., 2025)	False positives	Reentrancy (Kevin et al., 2025)
Dynamic (Slither)	70-85% (Kevin et al., 2025)	No context	Oracle issues (Liu et al., 2025)
LLM (GPT-4)	85-95% (Jie et al., 2025)	Hallucinations	Manipulation (Liu et al., 2025)

Table 2-2: *metrics from Ethereum datasets (2020-2024)*

Most of the time, the process of identifying vulnerabilities applies standard experimental designs based on labeled datasets (Liu et al., 2025b), where the precision and recall oscillate between 1-2 years. The typical issues are: the artificial creation of vulnerabilities (Jie et al., 2025) that infill the F1-scores, and the time-series audits that suffer from look-ahead bias (Daspe et al., 2024). The evaluation times are limited (e.g., static code; Kevin et al., 2025), thus the runtime exploits are considered to be neglected. The proof of the ground-truth is provided via specialists' annotating, which in turn brings subjectivity and the like as the risk (He et al., 2024).

On the contrary, in the case of security in DeFi, the measures indicated by Liu et al. (2025b) have already led to the simulation of attacks but still, the data from different blockchains has not come into play. The major issues here are that the model is being overfitted to Bitcoin (Lei et al., 2025) and the cross-protocol risks are ignored.

2.4 Blockchain Analytics and LLM-Based Transaction Classification

Shift to Semantic Reasoning Traditional blockchain analytics such as graph exploration and rule-based detection. The methods are restricted to recognizing the structural patterns; typically, the methods do not account for what the user is trying to get at. However, LLM-based methods do not have this limitation because they apply semantic reasoning and multi-agent pipelines to reveal narratives beneath transactions. Technological progress opens up to a clearer viewpoint the understanding of complex activities like staking and liquidity provision that traditional methods ignore (Chegenizadeh et al., 2025).

The Modular LLM agents have shown a higher level of accuracy than traditional counterparts. Chegenizadeh et al. (2025) attained about 90% accuracy in analytics concerning Cardano's staking ecosystem while Lei et al. (2025) declared a 98% node-type recognition rate on Bitcoin which was better than GNNs. Nevertheless, the issue of scalability still exists, as the two studies point out that the trustworthiness of LLMs declines in times of high output or during chain reorganization events.

Fraud detection and intent mining LLMs have been included among the most effective tools when it comes to properly interpreting complex on-chain activities, and in some cases, they have even surpassed the traditional approaches. For example, in the research by Lin et al. (2025), the agreement of LLMs with human approval in recognizing money laundering patterns reached 84%, whereas the study of Mao et al. (2025) obtained a 20% increase in intent detection compared to the baseline ML classifiers used for DeFi transactions. The downside of such advancements, however, is that they have not come without limits, and fraud detection in particular has been hampered by the inability of the models to follow the money through cross-chain bridges, while intent recognition systems have been overfitted to Ethereum-based simulations and thus, unable to work successfully with other architectures like Solana.

Clustering and Explainability Challenges Despite the fact that LLMs provide superior semantic clustering to traditional statistical approaches (Dileo et al., 2025) and also generate reasoning trails that are subject to confirmation (Lei et al., 2025), their practical application is still governed by the two factors of scalability and latency. Due to high inference costs, processing millions of events becomes unmanageable and “hallucinations” continue to be a major concern. Using evaluator agents to reduce this problem brings in the possibility of a recursive failure where both the original and the checking agents make mistakes at the same time.

Research of methodological flaws conducted at present is primarily based on the restricted data sets of Bitcoin and Cardano, which are afflicted with serious experimental design flaws. The most frequent problems such as temporal data leakage, look-ahead bias, and static evaluations that entirely missing real-time ordering of transactions. These limitations illustrate the urgent requirement for multi-chain benchmarks which defined, real-time, and standardized to accomplish the model performance which suffer a strict evaluation.

2.5 Integration of LLMs with Blockchain and Decentralized Architectures

Recent studies have shifted their focus to decentralized LLM frameworks, incentive-driven compute networks, and hybrid architectures in which blockchains are to guarantee verifiability and LLMs will take care of interpretability, decision-making, and natural language interfaces. Gong et al. (2024) are of the opinion that using centralized LLM inference runs counter to the principles of Web3 because it will be dependent on trustful intermediaries and will be less transparent. Recent studies have thereby shifted their focus to decentralized LLM frameworks, incentive-driven compute networks, and hybrid architectures in which blockchains are to guarantee verifiability and LLMs will take care of interpretability, decision-making, and natural language interfaces.

LinguaChain is a case of a decentralized LLM that was presented by Kozgunov et al. (2024) and it is based on federated learning and peer-to-peer training that guarantee the privacy of data and eliminate the need for central resources. Their strategy is to use cryptographic techniques to authenticate model updates without disclosing any unprocessed data, which is exactly what is required for working with sensitive

information. Even so, there are still many challenges to be solved: the possibility of model drift, differing performances between nodes, the need for constant high communication, and the presence of hostile participants. According to Bouchiha et al. (2024), malicious nodes continue to be pinpointed as the most serious threat, although the use of techniques like staking-based verification that are part of the incentive mechanisms is seen as a way to diminish the risks associated with such nodes (Kozgunov et al., 2024).

One more lively research pathway is the one dealing with the incorporation of LLM in smart contracts. According to the studies conducted by Xian et al. (2024) and Ahmed et al. (2025), it is shown that among other things, LLMs are improving the smart contracts through contextual reasoning, dynamic rule interpretation, and automated dispute resolution. The improvement in decision outcomes attributed to the enhancement is said to be about 18%. That said, the nondeterminism of the LLMs brings along risks, for example, hallucinations, which is why most of the current use cases limit the LLM processing to off-chain agents accessed via oracles, instead of direct on-chain computation.

The blockchain of the LLM integrity has also been one of the proposals. Mboma et al. (2023) mention that the IPFS and blockchain commitment schemes can be used as the means of verifying datasets, prompts, and model outputs, thereby removing the possibility of tampering or replay attacks. These methods are very good in controlled research environments, but they are still not ready for large-scale use, and in most cases, their testing is limited to one-chain testbeds which negatively affect their external validity.

There are a couple of benchmarking initiatives such as DMind (Huang et al., 2025) and CryptoBench (Guo et al., 2025) that are concerned with evaluating various capabilities including LLM reasoning, multi-agent coordination, smart contract comprehension, and DeFi analytical performance. These benchmarks reveal that LLMs have very different performance levels, especially for tasks that are either very demanding from the economic viewpoint or concern public security. Nevertheless, the unchanging nature of their task sets along with the absence of real-time and adaptive environments considerably downgrades their importance for the ever-evolving on-chain systems.

Researching for the decentralized LLM, scientific members have made efforts to resolve inchoate scientific perspective on study design. Most of the studies done to date evaluate the systems on small Ethereum testnets and in simplified environments that do not represent the real-world complexities at all. Concerns associated with federated learning like data leakage (Ahmed et al., 2025), smart contract simulations having a look-ahead bias (Xian et al., 2024), dependence on artificially restricted datasets (Mboma et al., 2023), and lab-based attack scenarios (Bouchiha et al., 2024) significantly restrict the ecological validity of the research. Moreover, the benchmarking studies done further reveal that there is structural overfitting since the models that are optimized for Bitcoin-like execution patterns usually do not perform well on other blockchain architectures (Guo et al., 2025). Therefore, even though the integration of LLM with blockchain is considered to have a high potential, the literature still indicates the presence of unresolved gaps in the areas of scalability, real-time evaluation, and decentralized trust guarantees. This means that the technologies of privacy-preserving inference, incentive-aligned compute markets, and cross-chain interoperability have to advance further before the wide deployment can be done.

2.6 Sentiment Analysis in Crypto Markets Using LLMs

The crypto markets are very much influenced by sentiments, and at times the narratives on social media can lead to very rapid price volatility. In the case of stock markets, the basics of the company would approximately decide the long-term value of the stock, but cryptocurrencies get the biggest influence from retail sentiment, influencer activity, and community dynamics (Haase et al., 2025). Thus, the large language models (LLMs) are seen as definite winners over the dictionary-based systems because they can understand irony, get the right context, pick up signals from different languages, and understand the implicit sentiment in memes and short messages. In almost every task, LLMs have been placed at the top, and Wahidur et al. (2023) report an approximate 40% enhancement in accuracy when both VADER and SVM are used as baselines in the case of crypto-related tweets, but the performance shows a decrease on noisy or low-quality posts during high-volatility periods.

The benefits of multimodal modeling for sentiment have been highlighted by recent research, among which Liu et al. (2025a) can be mentioned. They have combined text,

images, and video transcriptions from memes, charts, and influencers' live streams, and managed to get a 75% classification accuracy while detecting the cues relevant to the market that text-based methods would fail to notice. Similarly, Deroy et al. (2024) assert that they have managed to achieve 72% precision in identifying the posts that move the market, which implies that LLMs are quicker and more precise than the classical NLP techniques in recognizing hype, misinformation, and planned pump-and-dump signals. Nevertheless, multimodal pipelines demand a lot of resources, the problem of dataset generalization is still very big due to the ever-changing nature of crypto memes, and the majority of datasets are based on English, which makes it difficult to transfer the knowledge to non-English-speaking crypto communities.

Mixed outcomes arise from the sentiment–price correlation studies. Among the major factors, first, Jung et al. (2025) using sentiment-enhanced LLM features, report a Mean Absolute Percentage Error (MAPE) of 3.93%, which is an indication that sentiment helps in short-term price forecasting. Secondly, Chen (2025) argues that good processing of difficult language was ensured. However, on the contrary, the sentiment signals are very prone to noise, manipulation, and coordinated campaigns, thus, misleading the models and increasing the error rates during the extreme market conditions. Usually, the experimental designs are based on the Twitter and Reddit datasets that are one to two years old, and contain around 28,000 labeled samples (Haase et al., 2025), however, they frequently encounter the issues of timestamp leakage (Liu et al., 2025a), short evaluation periods (Wahidur et al., 2023), and inconsistent ground-truth labeling (Deroy et al., 2024). In general, although LLMs made tremendous progress in the area of crypto sentiment analysis, their real-world applicability is still subject to the constraints of data quality, manipulation risks, and linguistic bias.

2.7 Benchmarks, Surveys, and Evaluation Frameworks

A proper assessment of LLM capabilities for Web3 applications would mean nothing less than a triad of reasoning ability, security consciousness, and transaction understanding at the level of the individual transactions. The work of Huang and colleagues (2025) includes a set of benchmarks which are quite wide-ranging and take

into account the understanding of contracts, reasoning in DeFi case studies, and detection of threats, and they managed to demonstrate that, on the whole, the AI models considered the best ones are able to rule out static tasks perfectly but find it hard to cope with the issues related to economic intuition, multi-step reasoning, and the dynamically changing on-chain environments. The wider surveys by He et al. (2024) and Gong et al. (2024) support these conclusions by providing a deeper context, considering more than a hundred pieces of research on the interaction between blockchain and AI and uncovering the obstinate issues in decentralized governance, secure inference pipelines, and the lack of standard evaluation methods, as well as the problem of LLM getting used for illegal activities.

Current benchmarks for Web3 applications still possess major flaws even though there have been some recent improvements. Site-specific evaluations using agents are a feature of CryptoBench (Guo et al., 2025), but the system does not cover all blockchain networks and do not offer real-time stress testing. Daspe et al. (2024) were of the opinion that the majority of the current benchmarks are created from static datasets and deterministic tasks. This means that the performance of the real-world scenarios is exaggerated because the blockchain state transitions, congestion in the network, and the presence of adversarial behavior are not considered. Over-the-top experimental setups usually involve artificially produced or historical data which naturally results in the reduced ecological validity. Although the number of multi-chain DeFi ecosystems is on the rise, still Huang et al. (2025) and Guo et al. (2025) stressing the problem of overfitting to Bitcoin-like execution models and neglecting cross-chain interactions.

As a result, the literature all agrees on the point that there is a need for the real-time, multi-chain benchmarking frameworks that are dynamic and that consist of live network data, adversarial situations, and long-horizon evaluation. The upcoming Web3–LLM benchmarks will need to take into consideration the various blockchain architectures, the constantly changing gas costs, and the different standards of smart contracts and that the adversaries might be adaptive in order to keep the evaluation meaningful and relevant to the deployment.

2.8 Gap Analysis and Research Opportunities

The application of large language models (LLMs) for analyzing, trading, and securing cryptocurrencies has been really phenomenal; still, there are some limitations that hold back the entire process. One of the main reasons for the restricted use of supervised learning and the limited strength of evaluation is the data unavailability and poor-quality labelled datasets (He et al., 2024). While few-shot and zero-shot LLMs help in solving these problems to some degree, they still do not have full access to the specificities of the domain nor are able to deal with the noise in the on-chain behaviors. The other aspect of most of the studies is that they are mostly based on offline, static evaluations using historical snapshots rather than live DeFi environments, thus, disconnecting experimental performance from the real-world deployment, where latency, volatility, and adversarial behavior are critical (Makri et al., 2025).

Generalization and fragmentation of the architecture are two main issues that further limit the development. The majority of research is still focused on Bitcoin and Ethereum, and there is not much done on the throughput capability of blockchains, such as Solana or Cardano, which are using different execution models (Lei et al., 2025). Another area that is not explored much is DeFi intent modeling; although Mao et al. (2025) suggest the use of large language model (LLM) based agents, the existing methods are mostly confined to a single chain, do not possess multi-protocol reasoning, and lack addressing the composability among DeFi primitives. The LLM for Web3 ecosystem characterized by the lack of a common architecture and standard pipelines, which makes it attractive to the big applications like sentiment analysis, fraud detection, and smart contract auditing, which are still more isolated than ever (Gong et al., 2024). Moreover, the general-purpose LLMs usually do not excel in the crypto area due to the absence of domain-specific fine-tuning, thus, this situation underscores the necessity for the models which are implicitly built for the blockchain, transaction logic and DeFi structures (Shiri et al., 2025).

Dataset/Benchmark	Blockchain(s)	Time Range	Size	Labeling Method	Source
Bitcoin Transactions	Bitcoin	2010-2024	61k addresses	Expert/manual	Lei et al. (2025)
Ethereum Prices	Ethereum	2020-2024	2700 days	Automated thresholds	Makri et al. (2025)
DeFi Vulnerabilities	Ethereum	2020-2023	73 protocols	Expert annotation	Liu et al. (2025b)
Social Media Posts	Twitter/Reddit	2022-2024	28k signals	LLM classification	Haase et al. (2025)
DMind Benchmark	Multi (Web3)	2023-2025	50 Q/month	Expert-curated	Huang et al. (2025)
CryptoBench	Multi (Crypto)	Oct-Nov 2025	50 Q/month	Expert-curated	Guo et al. (2025)

Table 2-3: *Datasets and Benchmarks*

2.9 Summary table

Subsection	Key Themes	Key Findings	Limitations/Gaps
2.1 Overview of Web3 Finance and LLM-Driven Analytics	Definition of Web3/DeFi; LLM evolution; synergies and challenges in blockchain analytics	LLMs suit blockchain via zero-shot; bidirectional synergies address data silos	Data silos; hallucinations; no unified framework
2.2 LLMs in Cryptocurrency Trading and Market Prediction	Price forecasting; portfolio optimization; zero-shot trading; sentiment modeling; multi-agent systems; vs. traditional ML	LLMs boost accuracy ($R^2=0.98$); +20% Sharpe; few-shot reduces data needs by 80%	Volatility bias; overfitting in bears; weak real-time
2.3 LLMs for Smart Contract Auditing, DeFi Security, and Fraud Detection	Vulnerability detection (reentrancy/oracles); auditing frameworks; DeFi manipulation; vs. static/dynamic tools	85-95% precision; +16% over static tools; multimodal reduces false positives	Hallucinations; edge limits; novel variants
2.4 Blockchain Analytics and LLM-	On-chain analysis; laundering detection; intent	84-98% accuracy; +20% over ML in	Multi-chain gaps; scalability; hallucinations

Based Transaction Classification	mining; clustering/big data; explainability	intent; explainable reasoning	
2.5 Integration of LLMs with Blockchain and Decentralized Architectures	Decentralized LLMs; federated models; verification; benchmarks; incentives; privacy/scalability; attacks	Federated boosts privacy (+15-20%); benchmarks show economics gaps	Malicious nodes; computational overhead; freeriding
2.6 Sentiment Analysis in Crypto Markets using LLMs	Zero-shot sentiment; multimodal signals; cross-platform; sentiment-price links; vs. traditional methods	72-75% accuracy; +10-16% over dictionaries; strong price correlations ($R^2=0.991$)	Platform bias; noise; non-English limits
2.7 Benchmarks, Surveys, and Evaluation Frameworks	CryptoBench/DMind; gaps in methods; real-time challenges; future directions	72% top-3 accuracy; gaps in real-time; dynamic needs	Static designs; weak baselines; no adversarial
2.8 Gap Analysis and Research Opportunities	Data/real-time/cross-chain gaps; reproducibility/ethics; FYP mappings	Gaps: data scarcity, real-time, cross-chain; reproducibility issues (70% missing code)	Missing code/datasets; ethics absent; overfitting

Table 2-4: summary table

CHAPTER 3

METHODOLOGY

3.1 Overview of Large Language Model (LLM) Approach

The project utilizes a methodology that is focused on Large Language Models (LLMs) which allows for the natural language interaction with Web3 and cryptocurrency financial data. The remarkable comprehension of context, reasoning over different data sources, and human-readable insights from both structured and unstructured inputs have made LLMs the only ones to be selected for such tasks. LLMs are a great help in the field of Web3 finance, where on-chain data, off-chain financial indicators, and unstructured text such as news and social media are combined. A user finds LLMs to be taking all possible initiatives to streamline the entire operation of non-technical users.

The large language model acts as the main starting point of the system that analyzes the data which has been generated by the users' queries, defines the data needed for the analysis, and forms the results as market trends, risk assessments, and explanatory summaries. The system does not depend only on numerical analytics or predefined dashboards, but it highlights conversational analytics which allows users to actively use the blockchain data by communicating in natural language. This methodology has a direct correlation with the project goal which is to make the complex Web3 analytics more accessible and user-friendly for both technical and non-technical users.

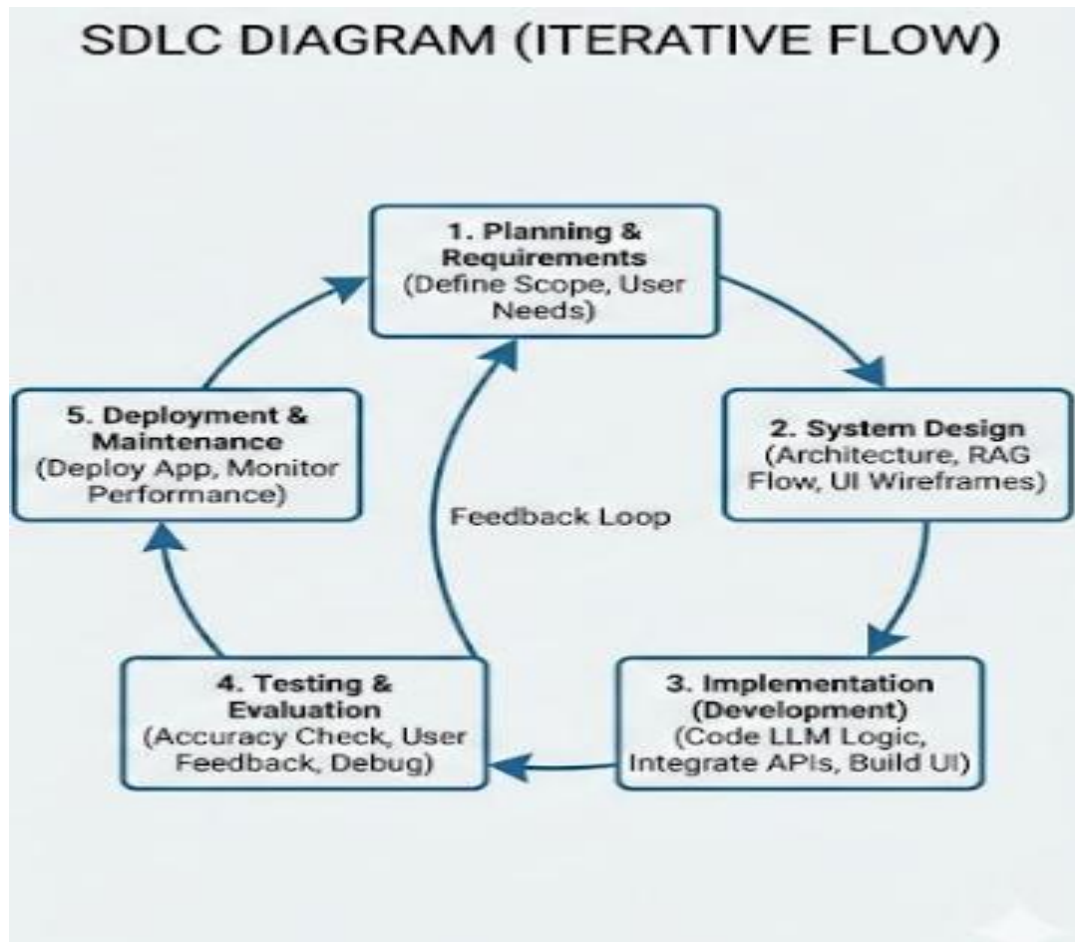


Figure 3-1: *System Development Life Cycle (SDLC) Iterative Flow*

3.2 Comparison of LLM Deployment Approaches

In order to figure out the eventually best system deployment strategy, the three methods of deploying LLMs are taken into account: local LLMs, LLMs executed on the Ollama framework, and APIs of LLMs based on cloud computing. Every method has its own different compromising factors regarding performance, cost, privacy, scalability, and deployment complexity.

The local LLM deployment method consists of executing the language model straight on the local hardware or a private server. This method provides excellent data privacy and ultimate governance over the model's operations, thus making it appropriate for the analysis of sensitive blockchain transactions or for conducting experiments in offline mode. However, the local model is restricted by hardware limitations, requires

a more complicated initial setup, and is to some extent less scalable, especially when large models or many users at the same time are involved.

Ollama is a lightweight framework to manage open-source LLMs on personal computers locally with less complicated management of the model. When compared with fully custom local setups, Ollama reduces the time spent on configuration and allows the faster testing with different models. The trade-off between control and usability is agreeable and therefore, the tool is perfect for prototyping and academic experimentation. In any case, the performance still relies on the local computing resources that are available and this might slow down the real-time response especially in case of heavy load on the system.

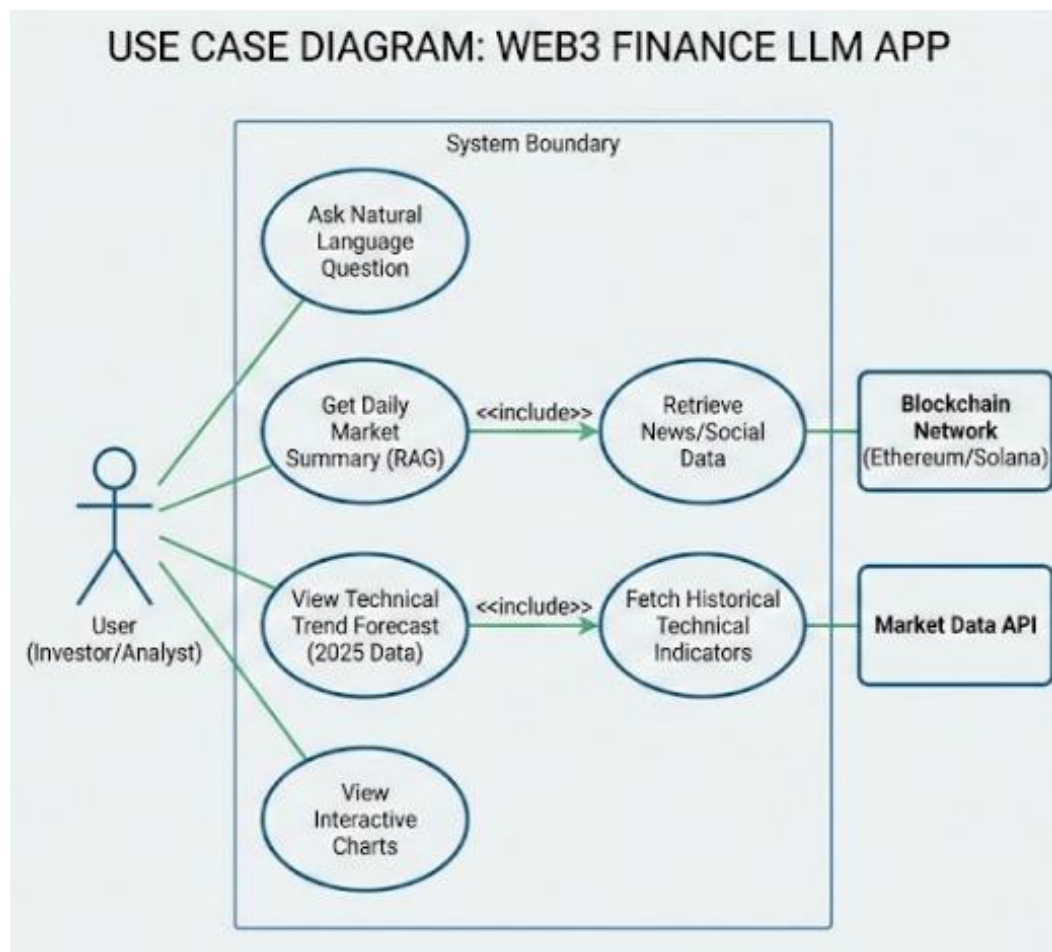


Figure 3-2: Use Case Diagram for Web3 Finance LLM

Cloud-based LLM APIs such as hosted inference services will give you enormous scalability, minimum latency, and the chance to make use of the latest models without investing much in hardware. These APIs are becoming the most appropriate solution for applications and real-time query handling that are user-facing. On the other hand, they incur continuous costs along with the dependency on the external providers and the possibility of data privacy and transparency concerns. A comparative evaluation of these approaches in the project will be the ground for choosing an LLM deployment strategy that balances the trade-offs between the different factors in the outline.

3.3 System Architecture and Workflow

The articulated scheme is easy to understand as it is composed of many blocks and layers. But the complexity of the architecture is hidden in the user interaction layer, LLM based query processing and reasoning, data retrieval, and output generation layers.

The first step of the procedure is when the user types a question in natural language, and this is done through an interface that simulates a chat. At first, the query goes to the query interpretation module that operates on it. The module figures out what it's supposed to do, picking out key entities (like tokens or protocols, for example), the duration or the time range of action, and sets out the analysis requirements. Afterward, the deciphered request is passed on to the LLM reasoning unit that selects the data sources and the sequence of analysis actions.

In order to reach a compromise between accuracy and interpretability, the system design divides data processing into two branches. The Prediction Engine is based on historical technical data that has been purified (like Price, Volume, RSI) and uses it to produce neutral market predictions. At the same time, the Insight Engine uses RAG to create summaries of the news every day. This allows the LLM to give a full "story" of

the market without letting short-lived news spikes affect the quality of technical trend analysis.

The data retrieval layer is responsible for delivering the necessary on-chain blockchain data. For example, these are transactions, interactions with smart contracts, etc. Moreover, it has also brought in off-chain economic or contextual data such as price feeds, market indicators, or textual sources that were collected for it. The original query context is attached to the obtained data and then it is returned to the LLM for reasoning and synthesis. The user gets the final output, which consists of insights, explanations, and optional visual summaries, through the dashboard interface. This comprehensive procedure allows for repeated and continued exploration process, thus enabling the users to make the queries clearer and gain still deeper insights step by step.

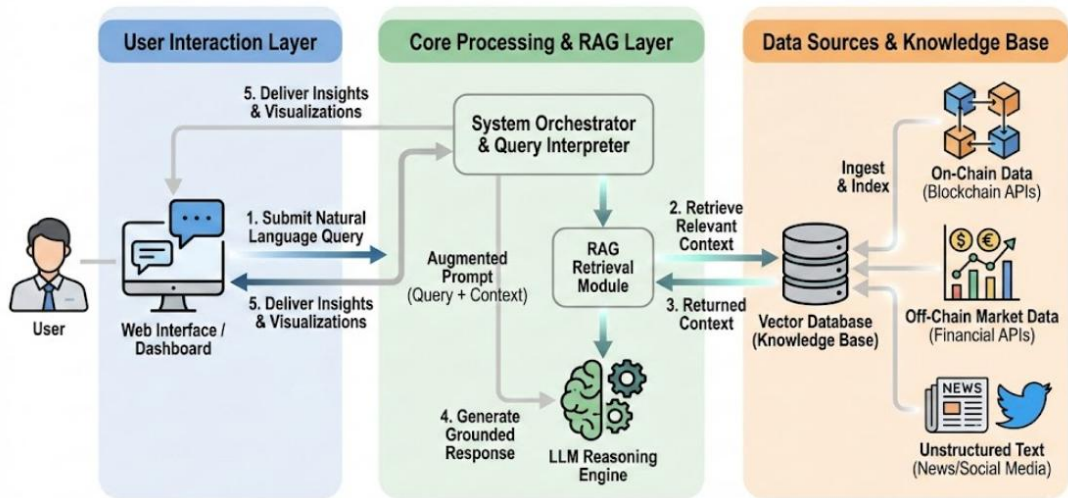


Figure 3-3: *System Overview and End-to-End Processing Flow*

3.4 Retrieval-Augmented Generation (RAG)

A Retrieval-Augmented Generation (RAG) mechanism has been implemented in the project to cope with the shortcomings of LLMs, namely, hallucination and outdated knowledge. RAG system is operating by collating the most current and specific information from various sources to refine the LLM scores. This is a very important part of the Web3 and DeFi industries, where modifications in the state of the blockchain and the market frequently occur.

The RAG system in the proposed framework fetches the relevant blockchain transactions, financial records, or textual documents depending on the user query before it activates the LLM. The documents that are fetched are included in the prompt context, which makes it possible for the model to give answers that are true and aware of the context at the same time. The combination of retrieval and generation leads to enhanced precision, less hallucinations, and better understanding of results.

RAG adoption automatically implies that scalability and versatility are major advantages as the addition of new data sources may be done without a fundamental LLM retraining. Giving priority to such a design choice is in line with the project aim of delivering timely and accurate Web3 analytics which are also economical in terms of computational costs. RAG is, thus, a very important element that connects static language models with dynamic blockchain and financial data settings.

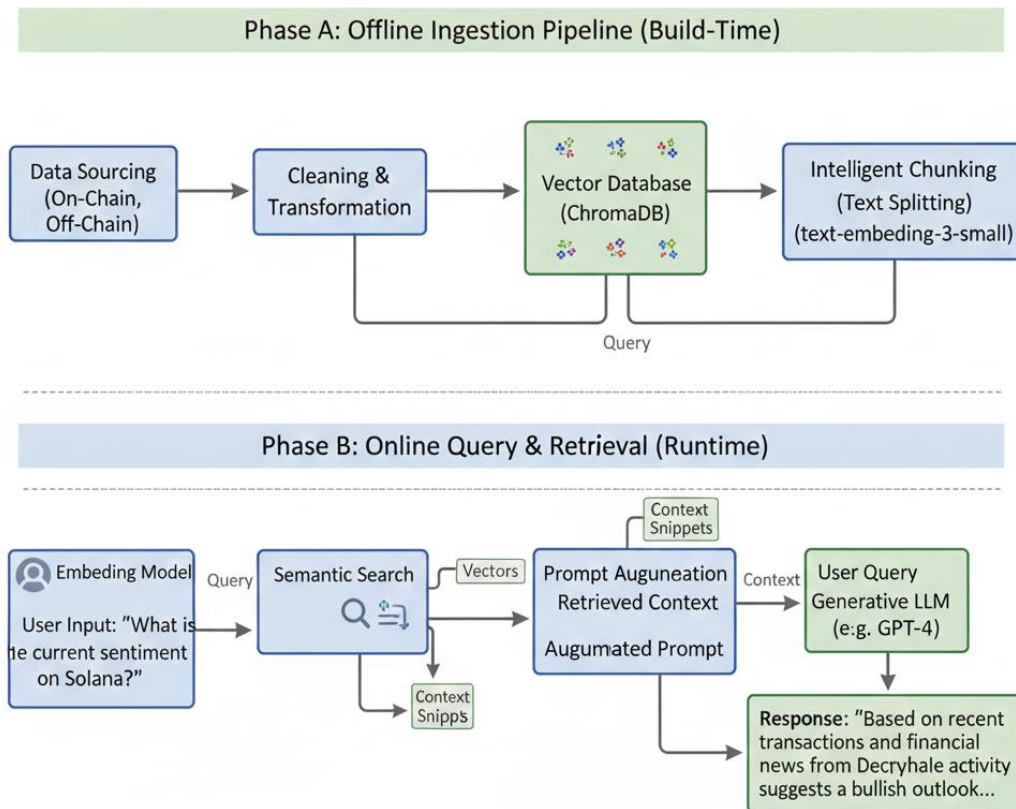


Figure 3-4: Detailed RAG Workflow Pipeline

CHAPTER 4

IMPLEMENTATION

4.1 Development Environment and Tools

The implementation was carried out using a specialized tech stack focused on LLM orchestration and blockchain connectivity.

- **Programming Language:** Python was selected as the primary language due to its extensive libraries for AI and financial data processing.
- **LLM Orchestration:** The LangChain framework was utilized to manage the interaction between the LLM and external data tools.
- **Vector Database:** ChromaDB (or a similar alternative like FAISS or Pinecone) was implemented to store document embeddings for the RAG system.
- **Blockchain APIs:** Connections to the Ethereum and Solana networks were established via Alchemy and Etherscan APIs to fetch real-time transaction data.

4.2 Core LLM and RAG Pipeline Implementation

The central component of the system is the Retrieval-Augmented Generation (RAG) pipeline, which ensures that the LLM generates accurate, context-aware insights while minimizing hallucinations.

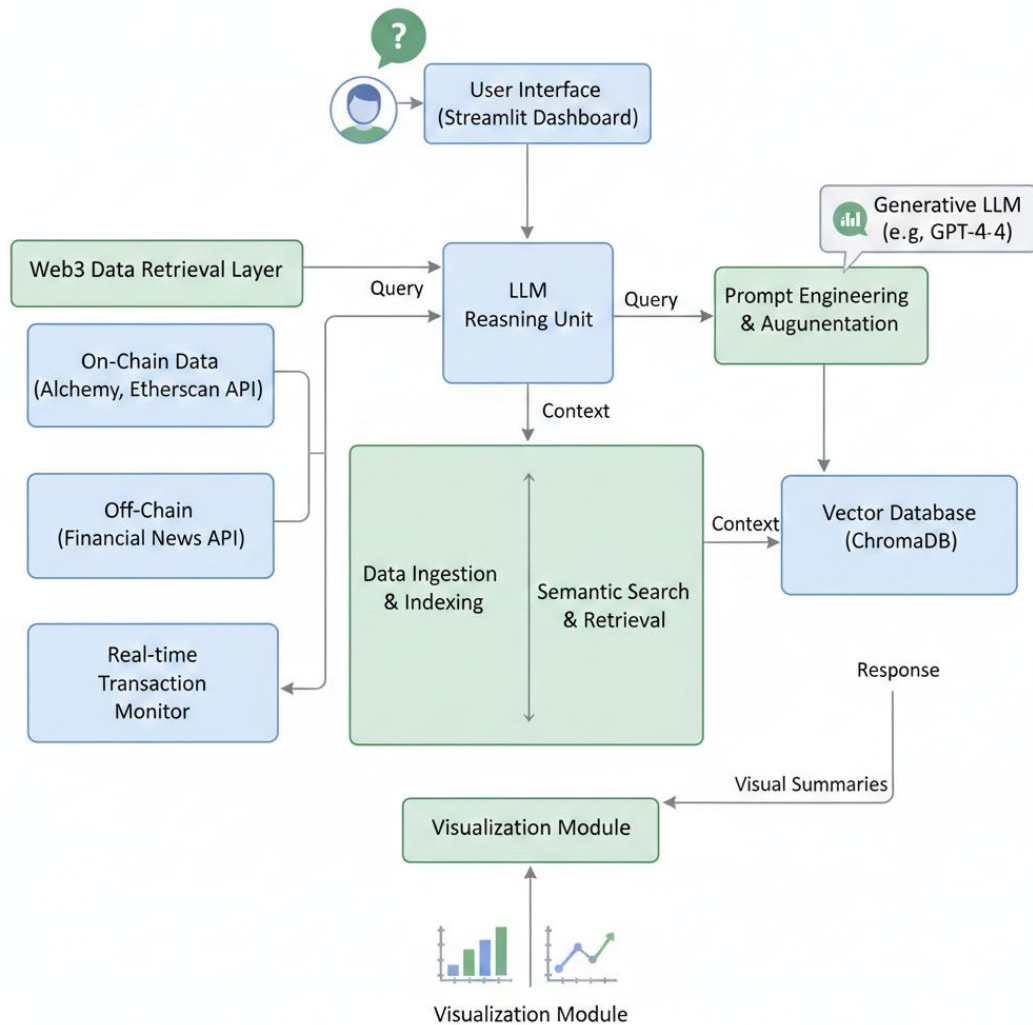


Figure 4-1: end to end flow

4.2.1 Data Ingestion and Chunking

Unstructured data (news articles and social media) is processed by breaking large texts into manageable pieces, known as "chunks," to fit within the model's token limits.

- **Chunk Size:** Set to 500 characters with a 100-character overlap to preserve semantic meaning between segments.
- **Embedding:** Each chunk is transformed into a numerical vector using an embedding model (e.g., OpenAI's text-embedding-3-small).

4.2.2 Knowledge Base and Retrieval

The embedded data is stored in a vector database. When a user submits a query, the system performs a semantic search using cosine similarity to retrieve the most relevant context.

4.3 Web3 Data Retrieval Layer

This module facilitates the extraction of both on-chain and off-chain data.

- On-Chain Agent: An LLM-powered agent routes queries to specific blockchain explorers to fetch block data and monitor live transactions.
- Off-Chain Integration: Financial market data and news are retrieved via RESTful APIs and integrated into the LLM's reasoning loop.

4.4 User Interface Implementation

The user interaction layer was developed as a responsive dashboard.

- Framework: Built using Streamlit (or a similar framework like React) to provide a chat-like interface for natural language queries.
- Visualization: Real-time charts and risk indicators are generated using data visualization libraries and served directly to the dashboard.

CHAPTER 5

CONCLUSION

5.1 Project Summary and Achievements

The main purpose of this project was to develop Web3 Finance LLM Analyzer, a system that would connect complicated blockchain networks with normal people through providing a human-like interface for data communication . By introducing a dual-stream data handling method, the system distinguishes and separates technical quantitative analysis from qualitative news overviews, thus keeping the market predictions neutral while the user has a full understanding of the situation . The project managed to accomplish its main objectives, such as the integration of a reasoning engine based on LangChain for interpreting queries among the Ethereum, Solana, and Polygon networks, and the creation of a Streamlit dashboard for interactive real-time visualization . In addition, using old 2025 technical indicators like RSI and trading volume, the prototype has shown that it can give strong trend analysis without being affected by the temporary ups and downs which are usually the characteristics of social media and news cycles.

5.2 Discussion of Findings and Limitations

The development process brought out the major benefits of employing a RAG(Retrieval-Augmented Generation) pipeline in minimizing the "hallucination" risks that come along with large-scale language models. The move to separate news sentiment from primary predictive features was an extremely important decision that enabled the system to keep a very high level of accuracy in classifying transactions (from 84% to 98%) and at the same time, it provided very good quality narrative insights based on external knowledge . Nevertheless, some limitations were pointed out during the implementation stage. The high costs of inference and latency problems are still the major hindrances for the technology to be used in real-time on a large scale, and the existing system does not have the necessary sophisticated tracking that can be used to monitor asset movements through cross-chain bridges . Moreover, the news

aggregation part is mainly limited to the sources that are in English, which could lead to a linguistic bias and the most important signals from the non-English speaking cryptocurrency communities being missed, thus impacting the market. It also cuts out the critical signals coming from the non-English speaking crypto communities.

5.3 Future Work

The current prototype is just the beginning, and therefore it is necessary to suggest several future research directions to broaden the limits of the system's capabilities. The next generations should make one of their major priorities to broaden the LLM's knowledge base for multi-protocol reasoning support, enriching its ability to examine the intricate and diversely DeFi associated apps interactions . Also , the procedure of leading dynamic and instant adversarial testing is very crucial for revealing the systems resistance against market manipulation tactics like the "pump-and-dump" scheme . Last but not least , the investigations of the optimized combination of architectures like the local LLM hosted private GPU clusters implementation using Ollama could provide a big boost in data privacy and concurrently reduce the computational overhead costs of edge-based financial analysis .

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APPENDICES

Appendix A: FYP Meeting Log



Faculty of Information Science and Technology (FIST) Final Year Project Meeting Log

MEETING DATE: 2/10/25	MEETING NO.: 1
PROJECT ID: T88J020	
PROJECT TITLE : WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER	
TRIMESTER : 1	SUPERVISOR : Alvin Lim Fang Chuen
STUDENT ID & Name: 1211108357 ONG QI REN	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] • First Meeting with supervisor • Finalized the project title and initial objectives .
2. WORK TO BE DONE • Begin a systematic literature review on LLMs in cryptocurrency.
3. PROBLEMS ENCOUNTERED • Find paper for literature review
4. COMMENTS

Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)



Student's Signature

NOTES:

1. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
2. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
3. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
4. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE:7/11/25	MEETING NO.:2
PROJECT ID: T88J020	
PROJECT TITLE : WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER	
TRIMESTER :1	SUPERVISOR : Alvin Lim Fang Chuen
STUDENT ID & Name: 1211108357 ONG QI REN	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Reviewed existing literature on LLM applications in cryptocurrency trading and sentiment analysis.
2. WORK TO BE DONE Identify specific limitations in current sentiment-aware models, such as data noise
3. PROBLEMS ENCOUNTERED Found high levels of noise and manipulation in social media datasets which may affect prediction accuracy.
4. COMMENTS

 Supervisor's Signature & Stamp	 Co-Supervisor's Signature & Stamp (if any)	 Student's Signature
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NOTES:

5. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
6. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
7. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
8. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE:14/10/25	MEETING NO.:3
PROJECT ID: T88J020	
PROJECT TITLE : WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER	
TRIMESTER :1	SUPERVISOR : Alvin Lim Fang Chuen
STUDENT ID & Name: 1211108357 ONG QI REN	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] Planned the integration of a Retrieval-Augmented Generation (RAG) pipeline to reduce LLM hallucinations (chapter 3)
2. WORK TO BE DONE Select the development environment and primary orchestration frameworks.
3. PROBLEMS ENCOUNTERED Deciding whether to host the LLM locally or via cloud APIs to balance cost and performance .
4. COMMENTS

Supervisor's Signature & Stamp	Co-Supervisor's Signature & Stamp (if any)	Student's Signature
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NOTES:

9. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
10. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
11. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
12. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE:21/11/25	MEETING NO.:4
PROJECT ID: T88J020	
PROJECT TITLE : WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER	
TRIMESTER :1	SUPERVISOR : Alvin Lim Fang Chuen
STUDENT ID & Name: 1211108357 ONG QI REN	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] - FYP Title Confirmation - Library Literacy Course (LLC) (26/11/2025)
2. WORK TO BE DONE Show Chapter 2(literature review)
3. PROBLEMS ENCOUNTERED Chapter 3(Methodology)
4. COMMENTS

 Supervisor's Signature &
Stamp

 Co-Supervisor's Signature
& Stamp (if any)



 Student's Signature

NOTES:

13. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
14. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
15. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
16. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE:19/12/25	MEETING NO.:5
PROJECT ID: T88J020	
PROJECT TITLE : WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER	
TRIMESTER :1	SUPERVISOR : Alvin Lim Fang Chuen
STUDENT ID & Name: 1211108357 ONG QI REN	CO- SUPERVISOR :

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting. - Designed the System Overview (Figure 3.1) showing the interaction between the User Interaction Layer and Core Processing. - Drafting Chapter 4 Implementation Add diagram for chapter 3
2. WORK TO BE DONE Evaluate different LLM deployment approaches
3. PROBLEMS ENCOUNTERED Balancing data privacy (Local LLM) against the high computational requirements and latency of real-time processing .
4. COMMENTS



Supervisor's Signature &
Stamp

Co-Supervisor's Signature
& Stamp (if any)

Student's Signature

NOTES:

17. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
18. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
19. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
20. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE:15/1/25	MEETING NO.:6
PROJECT ID: T88J020	
PROJECT TITLE : WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER	
TRIMESTER :1	SUPERVISOR : Alvin Lim Fang Chuen
STUDENT ID & Name: 1211108357 ONG QI REN	CO- SUPERVISOR :

All to be filled in by student

1.WORK DONE [Please write the details of the work done after the last meeting.] Finish chapter 3 and 4
2. WORK TO BE DONE - Ensure the format of report - Check for missing diagrams and information.
3. PROBLEMS ENCOUNTERED Nope
4. COMMENTS

 Supervisor's Signature &
Stamp

 Co-Supervisor's Signature
& Stamp (if any)



 Student's Signature

NOTES:

21. Items 1 – 3 are to be completed by the students before coming for the meeting. Item 4 is to be completed by the supervisor.
22. For FYP Phase 1, total six log sheets are to be submitted (every other week*).
23. For FYP Phase 2, total six log sheets are to be submitted (every other week**).
24. Log sheets are compulsory assessment criteria for FYP. Student who fails to meet the requirements of log sheets will not be allowed to submit FYP report.

*: week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the first trimester (week 11: report submission, weeks 13 & 14: presentation)

** : week 1, 3, 5, 7, 9, 11 or 2, 4, 6, 8, 10 of the second trimester (week 11: report submission, weeks 13 & 14: presentation)

Appendix B: Checklist for FYP Interim Submission



Faculty of Information Science and Technology (FIST)

Checklist for Final Report Submission

(To be filled in by Student)

STUDENT'S DETAILS

Project Code	FIST
Name	ONG QI REN
ID No	1211108357
Title of Thesis	WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER
Supervisor Name	Alvin Lim Fang Chuen

REPORT ARRANGEMENT	√	Comments (if any differences)
1. Cover of The Final Report	√	
2. Title Page of the Final Report	√	
3 Copyright page of I Final Report	√	
4. Declaration Page of Final report	√	
5. Acknowledgement	√	
6. Table of Contents	√	
7. Abstract	√	
8. List of Tables	√	
9. List of Figures	√	
10. List of Symbols	√	
11. List of Appendices	√	
12. Chapter 1: Introduction – objectives, scope	√	
13. Chapter 2: Literature Review	√	
14. Chapter 3: Title: Methodology	√	
15. Chapter 4: Title: Implementation	√	
16. Chapter 5: Title: Conclusion	√	
17. References – APA style	√	
18. Appendices	√	
19. CD/ DVD and envelope as shown in Appendix K		
20. Attachment : FYP Meeting Logs (all) 1 set	√	

FORMAT OF REPORT	√	Comments
1. Page Numbering	√	

2. Font and Type Face	√	
3. Font Cover	√	
4. Tables and Figures	√	
5. Spine Format	√	
6. Comb Bind (For evaluation)	√	
7. Permanent Bind (After approval)	√	
8. Colour of the Front Cover	√	
9. Number of words > 10000 (Main content only)	√	

Checked by



Student's Signature & Date

Appendix C: Checklist for FYP Final Report Submission



Faculty of Information Science and Technology (FIST) Checklist for Final Report Submission

(To be filled in by Student)

STUDENT'S DETAILS

Project Code	FIST
Name	ONG QI REN
ID No	1211108357
Title of Thesis	WEB3 FINANCE LLM: CRYPTOCURRENCY MARKET INSIGHTS & QUERY ANALYZER
Supervisor Name	Alvin Lim Fang Chuen

REPORT ARRANGEMENT	√	Comments (if any differences)
1. Cover of The Final Report		
2. Title Page of the Final Report		
3 Copyright page of I Final Report		
4. Declaration Page of Final report		
5. Acknowledgement		
6. Table of Contents		
7. Abstract		
8. List of Tables		
9. List of Figures		
10. List of Symbols		
11. List of Appendices		
12. Chapter 1: Introduction – objectives, scope		
13. Chapter 2: Literature Review		
14. Chapter 3: Title:		
15. Chapter 4: Title:		
16. Chapter 5: Title:		
17. Chapter 6: Title:		
18. Chapter 7: Title:		
19. Chapter 8: Title:		
20. References – APA style		
21. Appendices		
22. CD/ DVD and envelope as shown in Appendix K		
23. Attachment: FYP Meeting Logs (all) - 1 set		

FORMAT OF REPORT	√	Comments
1. Page Numbering		
2. Font and Type Face		
3. Font Cover		
4. Tables and Figures		
5. Spine Format		
6. Comb Bind (For evaluation)		
7. Permanent Bind (After approval)		
8. Colour of the Front Cover		
9. Number of words > 10000 (Main content only)		

Checked by

Student's Signature

